

Java OOP Assignment Solutions

Question 1:

Design a Java program to model a Bank (an abstract class) i) Having 2 data members bankName, branch and ii) Contains an abstract method getInterestRate() iii) Contains a concrete method display() to print a common message Create two subclasses: SBI and PNB. Subclasses have their own data members relevant to the corresponding classes. Each subclass must return a specific interest rate. In the main class create objects using reference of abstract class and implement run time polymorphism display bank details and interest rates.

Java Code:

```
abstract class Bank {
    String bankName;
    String branch;

    Bank(String bankName, String branch) {
        this.bankName = bankName;
        this.branch = branch;
    }

    abstract double getInterestRate();

    void display() {
        System.out.println("Welcome to Bank Management System");
    }
}

class SBI extends Bank {
    String managerName;

    SBI(String bankName, String branch, String managerName) {
        super(bankName, branch);
        this.managerName = managerName;
    }

    double getInterestRate() {
        return 6.5;
    }

    void showDetails() {
        System.out.println("Bank Name : " + bankName);
        System.out.println("Branch : " + branch);
        System.out.println("Manager : " + managerName);
        System.out.println("Interest Rate : " + getInterestRate() + "%");
    }
}

class PNB extends Bank {
    int noOfEmployees;

    PNB(String bankName, String branch, int noOfEmployees) {
        super(bankName, branch);
        this.noOfEmployees = noOfEmployees;
    }

    double getInterestRate() {
        return 7.0;
    }

    void showDetails() {
        System.out.println("Bank Name : " + bankName);
        System.out.println("Branch : " + branch);
        System.out.println("Employees : " + noOfEmployees);
        System.out.println("Interest Rate : " + getInterestRate() + "%");
    }
}

public class Main {
    public static void main(String args[]) {
```

```

        Bank b;

        b = new SBI("SBI", "Kolkata", "Rahul");

        b.display();
        ((SBI)b).showDetails();

        System.out.println();

        b = new PNB("PNB", "Delhi", 50);

        b.display();
        ((PNB)b).showDetails();
    }
}

```

Sample Output:

Sample Output:

```

Welcome to Bank Management System
Bank Name : SBI
Branch : Kolkata
Manager : Rahul
Interest Rate : 6.5%

```

```

Welcome to Bank Management System
Bank Name : PNB
Branch : Delhi
Employees : 50
Interest Rate : 7.0%

```

Question 2:

Create an interface “Department” containing attributes depName and depHead. It has abstract methods for getting data and printing the attributes and one default and one static method. Create a class “Hostel” containing attributes hostelName, hostelLocation and noOfRooms. Then write “Student” class extending “Hostel” class and implementing “Department” interface. Write suitable getdata and printdata methods for this class. Write a driver class to test the “Student” class. Print all the details of the student including depName and depHead.

Java Code:

```

interface Department {

    String depName = "CSE";
    String depHead = "Dr. Sen";

    void getData();
    void printData();

    default void showDepartment() {
        System.out.println("Department Function");
    }

    static void collegeName() {
        System.out.println("ABC Engineering College");
    }
}

class Hostel {

    String hostelName;
    String hostelLocation;
    int noOfRooms;

    void getHostelData(String hName, String hLoc, int rooms) {
        hostelName = hName;
        hostelLocation = hLoc;
        noOfRooms = rooms;
    }

    void printHostelData() {

```

```

        System.out.println("Hostel Name : " + hostelName);
        System.out.println("Hostel Location : " + hostelLocation);
        System.out.println("Number of Rooms : " + noOfRooms);
    }
}

class Student extends Hostel implements Department {
    String studentName;
    int regNo;
    String electiveSubject;
    double avgMarks;

    public void getData() {
        studentName = "Ankit";
        regNo = 101;
        electiveSubject = "AI";
        avgMarks = 85.5;

        getHostelData("Boys Hostel", "Kolkata", 120);
    }

    public void printData() {
        System.out.println("Student Name : " + studentName);
        System.out.println("Registration No : " + regNo);
        System.out.println("Elective Subject : " + electiveSubject);
        System.out.println("Average Marks : " + avgMarks);

        printHostelData();

        System.out.println("Department Name : " + depName);
        System.out.println("Department Head : " + depHead);
    }
}

public class Driver {
    public static void main(String args[]) {
        Student s = new Student();

        s.getData();

        Department.collegeName();

        s.showDepartment();

        s.printData();
    }
}

```

Sample Output:

Sample Output:

```

ABC Engineering College
Department Function
Student Name : Ankit
Registration No : 101
Elective Subject : AI
Average Marks : 85.5
Hostel Name : Boys Hostel
Hostel Location : Kolkata
Number of Rooms : 120
Department Name : CSE
Department Head : Dr. Sen

```

Question 3:

Create a package named "Employee" which has a class named Salaried with following variables: Emp_name, Emp_id, Age, Salary, Years of experience, Department name. Write Getdata and printdata methods for above class. Create a subclass of Salaried named "Trainee" in another file (not included in Employee package). Trainee has

following extra features: Training Domain, Date of joining, Training score. Write getdata() and printdata() methods and a method to print those trainee names who joined more than 1 year before and got a score above 80.

Java Code:

```
// Salaried.java
package Employee;

public class Salaried {

    public String emp_name;
    public int emp_id;
    public int age;
    public double salary;
    public int yearsExperience;
    public String departmentName;

    public void getdata(String name, int id, int a,
                        double sal, int exp, String dept) {

        emp_name = name;
        emp_id = id;
        age = a;
        salary = sal;
        yearsExperience = exp;
        departmentName = dept;
    }

    public void printdata() {

        System.out.println("Employee Name : " + emp_name);
        System.out.println("Employee ID : " + emp_id);
        System.out.println("Age : " + age);
        System.out.println("Salary : " + salary);
        System.out.println("Experience : " + yearsExperience);
        System.out.println("Department : " + departmentName);
    }
}

// Trainee.java
import Employee.Salaried;

public class Trainee extends Salaried {

    String trainingDomain;
    String dateOfJoining;
    int trainingScore;

    public void getdata(String name, int id, int age,
                        double salary, int exp, String dept,
                        String domain, String doj, int score) {

        super.getdata(name, id, age, salary, exp, dept);

        trainingDomain = domain;
        dateOfJoining = doj;
        trainingScore = score;
    }

    public void printdata() {

        super.printdata();

        System.out.println("Training Domain : " + trainingDomain);
        System.out.println("Date of Joining : " + dateOfJoining);
        System.out.println("Training Score : " + trainingScore);
    }

    public void checkEligibility() {

        if(yearsExperience > 1 && trainingScore > 80) {
            System.out.println(emp_name + " is eligible.");
        }
    }

    public static void main(String args[]) {
```

```
        Trainee t = new Trainee();

        t.getdata("Riya", 201, 24, 35000,
                  2, "IT", "Java",
                  "10-01-2024", 90);

        t.printdata();

        System.out.println();

        t.checkEligibility();
    }
}
```

Sample Output:

Sample Output:

Employee Name : Riya
Employee ID : 201
Age : 24
Salary : 35000.0
Experience : 2
Department : IT
Training Domain : Java
Date of Joining : 10-01-2024
Training Score : 90

Riya is eligible.